



TNX CODATHON 2K26

FINTECH + AI TRACK



Voice Khata

AI-Powered Voice-First Digital Ledger for Small Merchants

"Khata likhna. Ab bas bolkar." — Zero typing bookkeeping in seconds.

TEAM FIN FUSION

- Harshika Dixit : Team Leader | Research | Testing
- Avika Srivastava : UI/UX Design | AI Training | Testing
- Vansh Gupta : Backend Development
- Vineet Singh Chandel : Frontend + Backend Development



Problem Statement

Millions of Indian retail shopkeepers lose hours and money to manual paper Khatas.

01

Manual & Error - Prone

Handwritten notebook entries are hurried, messy, and cause math errors.

02

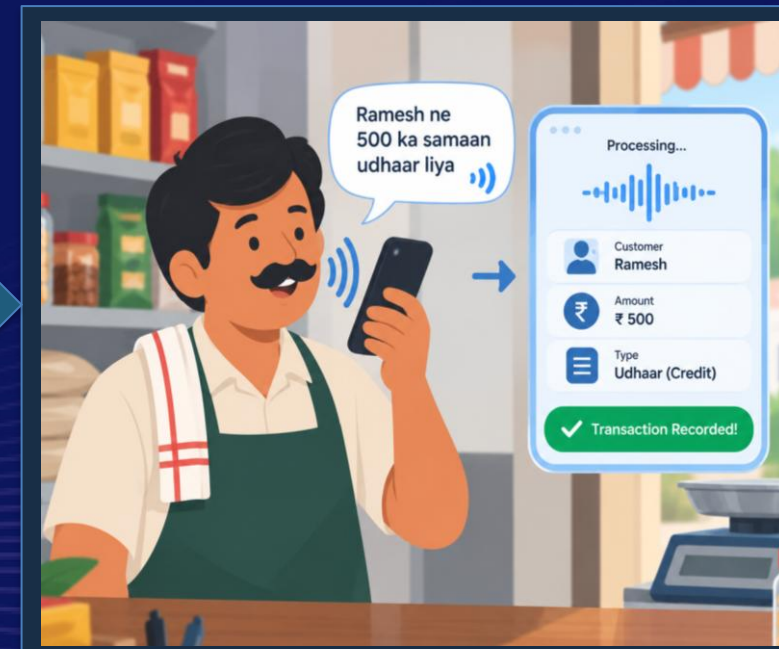
Counter Typing Friction

Shopkeepers cannot type on keyboards or navigate apps while serving customers.

03

Loss & Dispute Risks

Damaged, lost, or disputed notebooks lead to unrecoverable customer debt.



Our Goal: Make Digital Book keeping as natural as speaking

Proposed Solution

Zero typing. Spoken Hindi/English converted to verified digital records in 3 seconds.

Voice Capture

EN → अ

Business

Personal

ADD TRANSACTION

Speak → Review → Save

Say what happened in Hindi or English, review details, and save directly to your ledger.

← Back to overview

Voice-first capture

Review entry

REVIEW ENTRY Business Mode

Review before saving

Record again

Heard: "Ramesh ne ₹500 jama kiye"

+ ₹ Received / जमा मिला (Credit)

Payment from customer / Income

- ₹ Paid / उधार दिया (Debit)

Udhaar to customer / Expense

Parsed deterministically using VoiceKhata local ledger engine.

Customer / Party Name

₹ Amount (₹)

Ramesh

500

Category

Payment Method

Sales

UPI

Transaction Date

14 - 09 - 2026

Cancel

✓ Save with confidence

3-Second Flow · Zero Typing · 100% Audit Confidence



Natural Voice Input

Speak naturally in Hindi, English, or conversational Hinglish.



AI Entity & Intent Parser

Groq Llama + deterministic engine extracts party, amount & type.



1-Click Review & Confirm

Visual confirmation card verifies details before saving.



Instant Ledger Update

Updates customer running balance in Supabase cloud instantly.

Technology & Architecture



Frontend & PWA

- **Core Stack:** React 19 / 18, TypeScript, Vite 7
- **Styling:** Tailwind CSS 4, Radix UI Primitives
- **Visuals:** Chart.js, Recharts, Framer Motion
- **Mobility:** Progressive Web App with Service Workers

Backend & Cloud

- **Runtime:** Node.js & Express 5 (TypeScript)
- **Database:** Supabase PostgreSQL 15 with RLS
- **Auth:** Firebase Auth & Google OAuth 2.0
- **Automation:** Node-Cron scheduled reminders

AI & Speech Engine

- **Speech STT:** In-Browser Web Speech API (Hindi/English)
- **LLM Cloud:** Groq Cloud SDK (Meta Llama 3.1)
- **NLP Logic:** Deterministic regex & keyword parser
- **Vision OCR:** Google Cloud Vision OCR for bills



Implementation Roadmap

VoiceKhata was built in five structured phases — from merchant research to a fully functional live prototype — within the hackathon timeline.

01 — Problem Analysis

Understand merchant pain points through research. Identify gaps in existing digital ledger solutions. Define core requirements for voice-first design.

1

2

02 — Design

UI/UX design in Figma. User flow mapping for voice and text interactions. Wireframes validated against merchant mental models.

03 — Development

Parallel build of Frontend (Tailwind), Backend (FastAPI), Voice pipeline (Web Speech API + Whisper), and NLP transaction extraction engine.

3

4

04 — Testing

Voice accuracy validation across Hinglish inputs. Ledger consistency checks. Edge case handling for ambiguous transactions.

05 — MVP Launch

Working prototype deployed on Render. Live demo ready. Hackathon submission packaged with full feature documentation.

5

Expected Impact & Features

VoiceKhata targets a large and underserved segment of India's retail economy — merchants who manage credit transactions daily but have been excluded from digital financial tools due to literacy, language, and usability barriers.

Who We Serve



Kirana & Grocery Stores

High-volume, daily credit transactions with regular customers



Tea Stalls & Street Vendors

Hands-free recording essential during peak service hours



Pharmacies & Small Retail

Accuracy-critical ledger entries requiring fast, reliable logging



Any Merchant with a Khata

If they write it by hand today, VoiceKhata can digitize it

Business and Social Impacts

Faster Recording

Speak instead of write — transactions logged in seconds, not minutes

Fewer Manual Errors

AI extraction eliminates arithmetic mistakes and illegible entries

Instant Balance Tracking

Running balances calculated automatically after every transaction

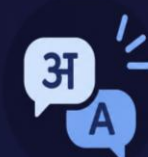
Operational Efficiency

Less time on bookkeeping, more attention on customers and sales



Financial Inclusion

Brings digital bookkeeping to merchants excluded by typing-first apps



Language Accessibility

Hindi-English support removes the language barrier to digital adoption



Future Scope & Improvements

Building on the MVP, VoiceKhata can evolve into a comprehensive AI-powered business assistant for India's small merchants.



Voice & Accessibility

- Support for more Indian languages
- Natural conversation & voice commands
- WhatsApp integration for voice entries



Business Growth

- Inventory & purchase management
- AI-powered insights and sales reports
- GST-ready invoicing and bills



Payments & Expansion

- UPI payment reminders and collections
- OCR bill/receipt scanning
- Offline mode for low-connectivity areas

How VoiceKhata Stands Out



Feature	 VoiceKhata	 Khatabook	 OkCredit	 Vyapar	 myBillBook
Voice-first entry	✓	—	—	—	—
Hindi / Hinglish voice input	✓	—	—	—	—
AI-based transaction extraction	✓	—	—	—	—
Digital customer ledger	✓	✓	✓	✓	✓
Credit & payment tracking	✓	✓	✓	✓	✓
Transaction history	✓	✓	✓	✓	✓
Business reports	✓	✓	✓	✓	✓
Billing & invoicing	◐	✓	◐	✓	✓
Inventory management	◐	✓	◐	✓	✓
GST / accounting tools	◐	✓	◐	✓	✓
Multi-device access	✓	✓	✓	✓	✓
Primary focus	Voice-first simplicity	Digital khata for small businesses	Digital ledger and credit management	Complete business accounting	Billing + accounting for growing businesses



= Core feature



= Limited / Expandable



= Not a core feature



Voice Khata Preview

1

Speak

Record a Transaction naturally

ADD TRANSACTION

Speak → Review → Save

Say what happened in Hindi or English, review details, and save directly to your ledger.

← Back to overview

Voice-first capture Step 1 of 1 - Review before saving

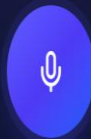


VOICE ENTRY

Say what happened

Speak naturally in Hindi, English, or Hinglish.

English (India) IN हिंदी / Hinglish



TRY SAYING / बोलकर देखें

"Ramesh ne ₹500 jama kiye"

2

Review

AI automatically extracts details

ADD TRANSACTION

Speak → Review → Save

Say what happened in Hindi or English, review details, and save directly to your ledger.

← Back to overview

Voice-first capture Review entry

REVIEW ENTRY Business Mode

Review before saving

Heard: "Ramesh ne ₹500 jama kiye"

Record again

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Udhaar to customer / Expense

Parsed deterministically using VoiceKhata local ledger engine.

Customer / Party Name

Ramesh

₹ Amount (₹)

500

Category

Sales

Payment Method

UPI

Transaction Date

14 - 09 - 2026

Cancel

Save with confidence

3

Track

View updated ledger and insights

Khata

Customers, suppliers, and balances in one place.

+ Add Customer / Supplier

Total to receive

₹4,500

3 From customers

Total to pay

₹10,050

To suppliers & wholesalers

Overdue

₹2,450

Requires payment follow-up

Q Search customer by name or phone...

All (5)

Customers Suppliers

Interactive Gestures: Swipe customer row Right to quickly record payment, or Left for instant WhatsApp reminder.

Drag with mouse or touchscreen

R Ramesh Kumar CUSTOMER

98765 43210 • Last activity: Today

₹2,450

You will receive

G Gupta Traders SUPPLIER

98123 76540 • Last activity: Today

₹6,850

You will pay

S Suresh Sharma CUSTOMER

98980 12345 • Last activity: Today

₹1,200

You will receive

Live Demo Visit the deployed prototype: [Voice khata](#)